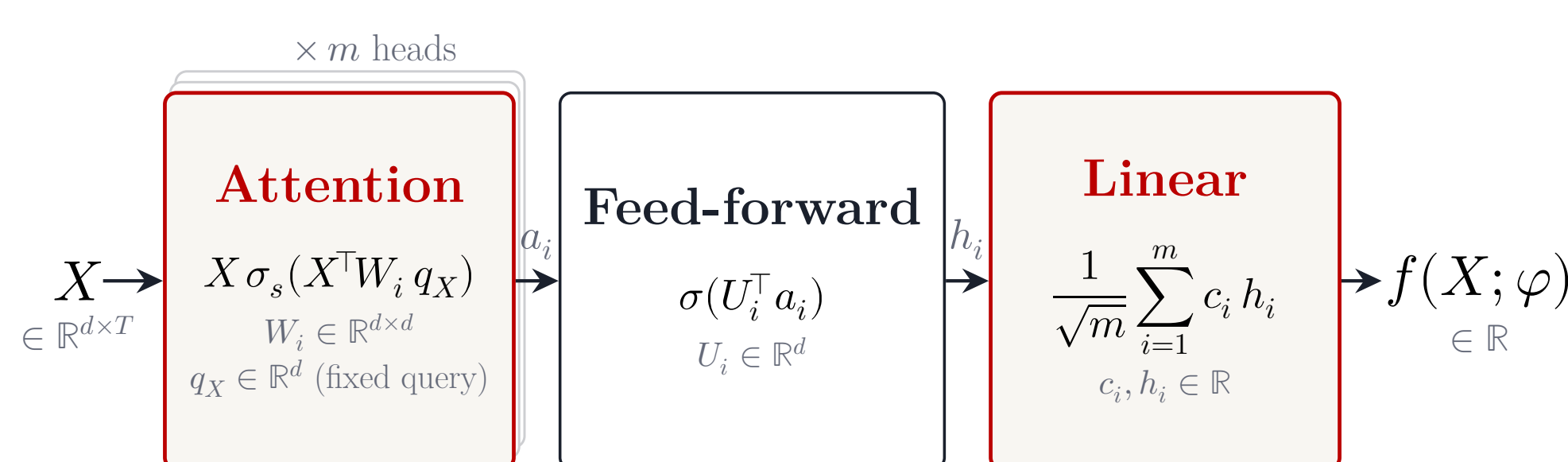


💡 Motivation

Modern AI runs on Transformers — but most of what we know about them is empirical. This work gives a **mathematical guarantee** for how they learn: training is stable at moderate width, and longer inputs do not slow training down; they only cost memory.

⚙️ Multi-Head Transformer



Assumptions: bounded tokens $\|X_t\|, \|q_X\| \leq 1$; bounded smooth σ .

📋 Problem & Algorithm

Seq-to-scalar regression — e.g. predicting a review's rating from text, or the next value of a time series.

Setup. n i.i.d. samples $\mathcal{D} = \{(X^{(j)}, y^{(j)})\}_{j=1}^n$, with $X^{(j)} \in \mathbb{R}^{d \times T}$, $y^{(j)} \in \mathbb{R}$. Mini-batch empirical MSE:

$$\widehat{\mathcal{L}}_B(\varphi) := \frac{1}{|B|} \sum_{j \in B} (f(X^{(j)}; \varphi) - y^{(j)})^2$$

Algorithm: Proj(S)GD. With step size η and mini-batch $B_s \subseteq [n]$,

$$\varphi^{(s+1)} = \Pi_{\Omega_\rho}(\varphi^{(s)} - \eta \nabla_\varphi \widehat{\mathcal{L}}_{B_s}(\varphi^{(s)}))$$

PROJGD : $B_s = [n]$

PROJSGD : $B_s = \{J_s\}$ with $J_s \stackrel{\text{i.i.d.}}{\sim} \text{Unif}([n])$.

🧩 Technical Challenges

- **Softmax attention** is highly nonlinear with cross-token coupling.
- **Non-convex** loss landscape.

How prior work copes (and limitations):

- **Linear attention:** drop σ_s entirely (loses nonlinearity)
- **Pointwise activation:** replace σ_s with ReLU (loses token coupling)
- **Infinite width with $\frac{1}{m} W_K^\top W_Q$:** softmax converges to mean-pooling
- **Positive-definite Gram matrix:** linear convergence rate, but requires $d \geq n$

★ Contributions

- **True softmax attention:** no surrogate used
- **No positive-definite Gram matrix** required
- **Logarithmic** width in sample size: $m \gtrsim \log(n)$
- **T -independent** empirical loss

🏠 Main Result

Theorem 1. Let $f^* \in \mathcal{F}_{\bar{\nu}}$ be the target function. After τ steps of PROJGD with an appropriate step size and projection radius, the average iterate $\bar{\varphi}^{(\tau)}$ satisfies w.h.p.:

$$\widehat{\mathcal{L}}_n(\bar{\varphi}^{(\tau)}) \lesssim \underbrace{\frac{D^2}{\sqrt{\tau}}}_{\text{optimization}} + \underbrace{\sqrt{\frac{D^3}{m}}}_{\text{linearization}} + \underbrace{\sqrt{\frac{D \log(n/\delta)}{m}}}_{\text{approximation}}$$

where $D := d + \log(m)$.

- **Non-asymptotic:** holds for any finite $\tau, m \geq 1$.
- Under PROJSGD, same bound holds for $\mathbb{E}[\widehat{\mathcal{L}}_n(\bar{\varphi}^{(\tau)}) | \varphi^{(0)}]$ with identical rates.

⚡ Key Ideas

- **Near-initialization regime.** Parameters stay in $\Omega_\rho \Rightarrow f$ is well-approximated by its linearization around init:
 $f_{\text{lin}}(X; \varphi) := f(X; \varphi^{(0)}) + \langle \nabla f(X; \varphi^{(0)}), \varphi - \varphi^{(0)} \rangle$

- **Transportation mappings.** Every target $f^* \in \mathcal{F}_{\bar{\nu}}$ has an associated function v^* — the *transport map* — that tells initial parameters how to shift:

$$\tilde{\varphi}_i = \varphi_i^{(0)} + m^{-1/2} v^*(\varphi_i^{(0)})$$

Transported parameters approximate f^* . The constraint $\|v^*\| \leq \bar{\nu}$ sets the regression problem's difficulty.

- **Softmax Jacobian = covariance.** Softmax Jacobian gives $\|X J_{\sigma_s} X^\top\|_{\text{op}} \leq 1$ uniformly in $T \Rightarrow h$ is L_1 -Lipschitz, L_2 -smooth on Ω_ρ with

$$L_1 \lesssim \underbrace{\sqrt{d + \log m}}_{\text{no } T \text{ dependence}}, \quad L_2 \lesssim d + \log m$$

RNN contrast: IndRNN gradient is a product of T Jacobians $\Rightarrow L_1, L_2 \sim \exp(T)$.

🔧 Transformer NTK

- **Definition.** Inner product of init-gradient features in the infinite-width limit:

$$K(X, X') := \lim_{m \rightarrow \infty} \langle \nabla_\varphi f(X; \varphi^{(0)}), \nabla_\varphi f(X'; \varphi^{(0)}) \rangle$$

- **Block decomposition.** Gradients split across (c, U, W) :

$$K = K_c + K_u + K_w$$

Non-degenerate attention kernel: $K_w \neq 0$ — unlike $\frac{1}{m} W_K^\top W_Q$ infinite-width limit where $K_w = 0$.

- **Transformer-NTK RKHS.** Transport maps v define functions of the form

$$\tilde{f}(X; v) := \mathbb{E}[\langle \nabla_\varphi f(X; \varphi_0), v(\varphi_0) \rangle]$$

which inherit an inner product from their transport maps:

$$\langle \tilde{f}(\cdot; v), \tilde{f}(\cdot; v') \rangle_{\mathcal{F}} := \mathbb{E}[\langle v, v' \rangle]$$

The inner product has the reproducing property w.r.t. the Transformer-NTK:

$$\langle \tilde{f}(\cdot; v), K(\cdot, X') \rangle_{\mathcal{F}} = \tilde{f}(X'; v)$$

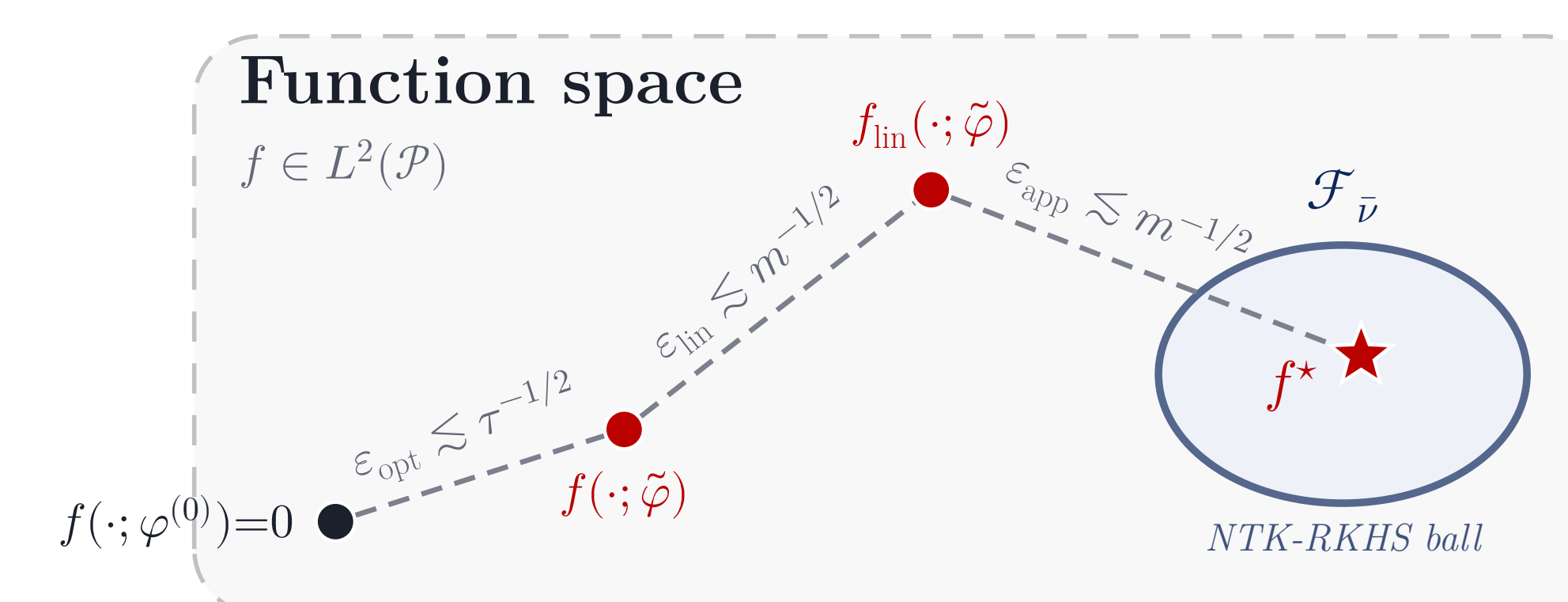
So completion $\bar{\mathcal{F}}$ is the Transformer-NTK RKHS!

- **Target class.** The target f^* lives in a norm-bounded ball inside $\bar{\mathcal{F}}$:

$$\mathcal{F}_{\bar{\nu}} := \{f \in \bar{\mathcal{F}} : \|f\|_{\mathcal{F}} \leq \bar{\nu}\}$$

$\bar{\nu}$ quantifies the regression problem's complexity.

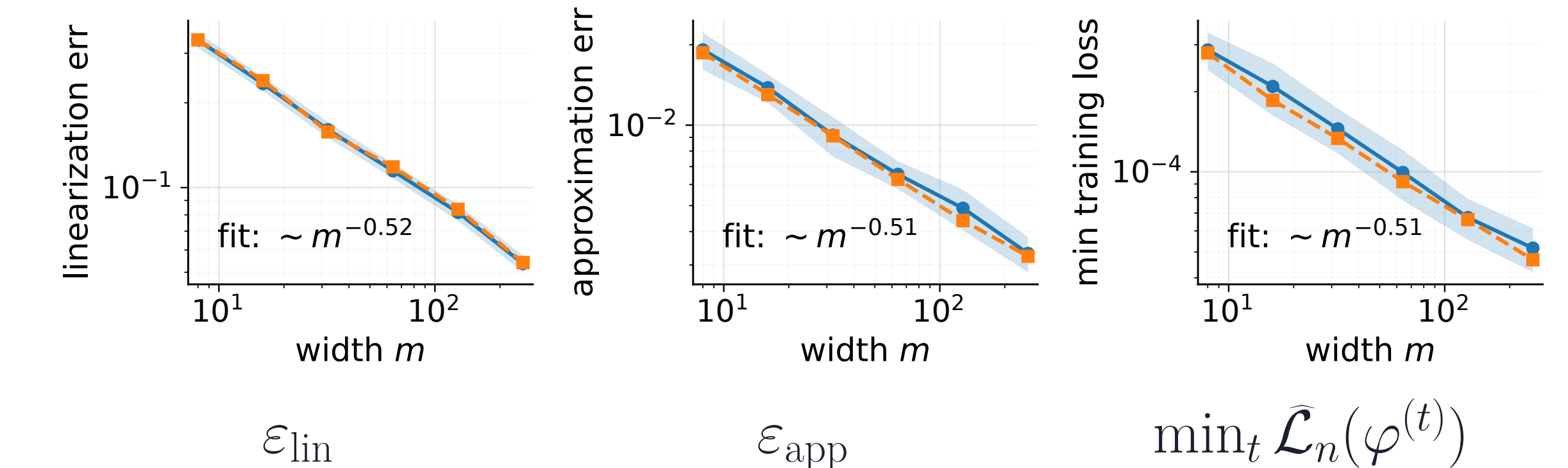
📦 Error Decomposition



Bound	Why
$\epsilon_{\text{lin}} \lesssim (L_1 + L_2) \rho^2 / \sqrt{m}$	Taylor remainder in Ω_ρ
$\epsilon_{\text{app}} \lesssim \bar{\nu} \sqrt{\log(n)/m}$	Concentration over m heads
$\epsilon_{\text{opt}} \lesssim D^2 / \sqrt{\tau}$	ProjGD Lyapunov

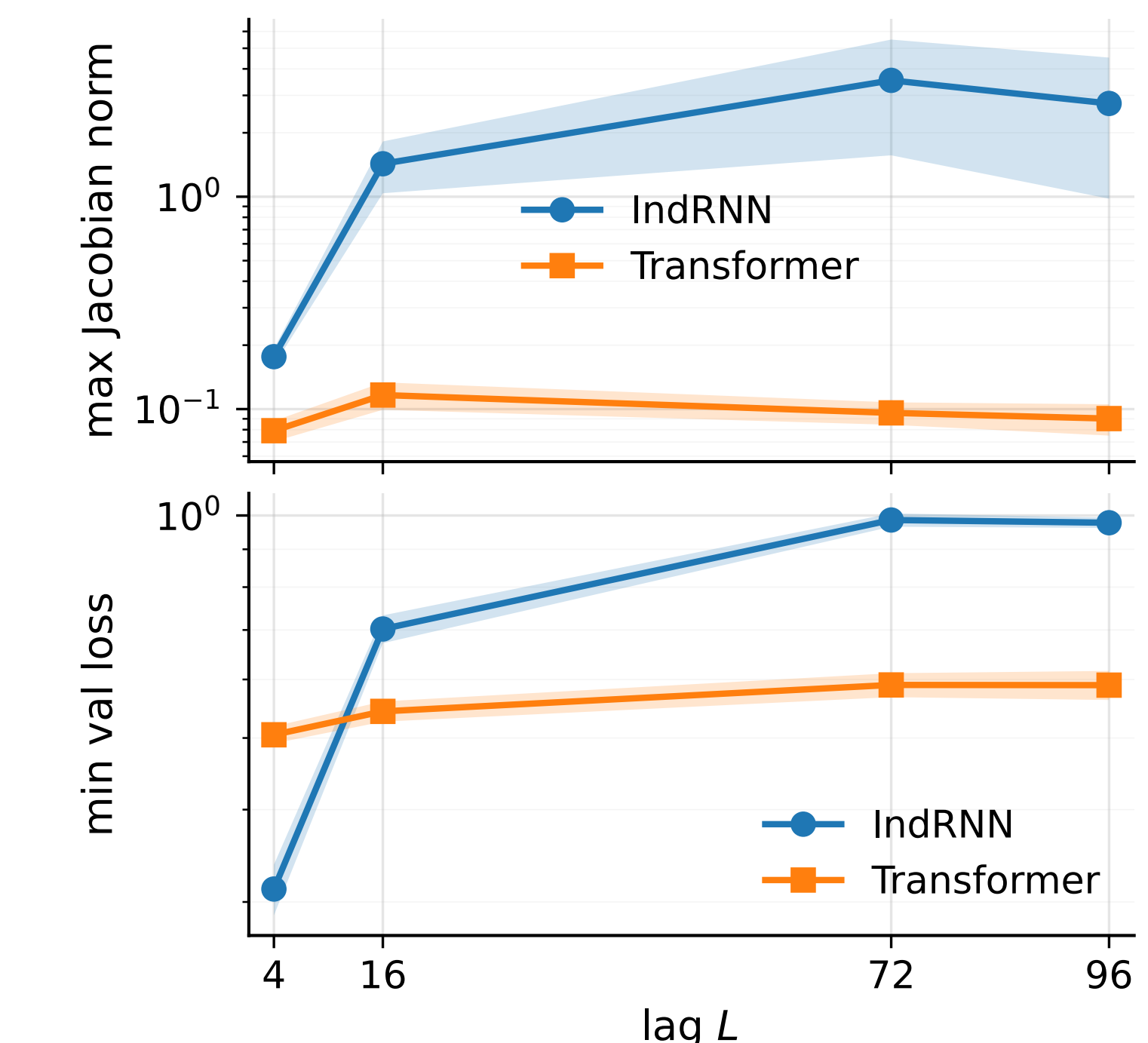
🔥 Width scaling validates theory

Linearization and approximation decay as $m^{-1/2}$; at fixed τ , the min training loss follows the same trend:



⚖️ Transformer vs. IndRNN

Setup. Synthetic regression on an autoregressive process: $X_t = 0.9X_{t-L} + \varepsilon_t$ where $\varepsilon_t \sim \mathcal{N}(0, 1)$. Input: $(X_1, \dots, X_T)$; output: next value $y = X_{T+1}$. Lag L sets the dependency horizon.



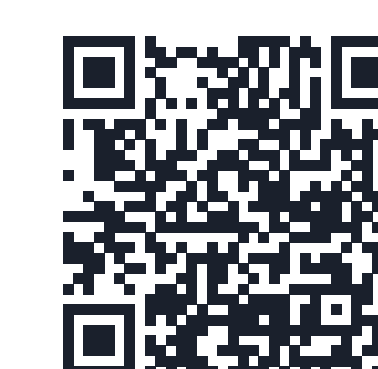
Top: training stability: max Jacobian norm.
Bottom: performance: min validation loss.

- IndRNN Jacobian grows with L ; Transformer stays stable.
- IndRNN loss degrades with L ; Transformer's is flat.
- **Trade-off:** memory $O(L)$ (Transformer) vs $O(1)$ (IndRNN).

✅ Takeaways

- Transformers do not need to be huge to train reliably — modest width is enough.
- Longer inputs do not destabilize training or width requirement — the softmax keeps things under control.
- The cost of handling long inputs is memory, not training time or width.

🔗 Code & Funding



github.com/enesarda22/nonasymptotic-transformer

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